

LUCKY LAKE, SK, Canada

WMO: 714550

Lat: 50.950N

Lon: 107.150W

Elev: 665

StdP: 93.59

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-28.3	-34.9	0.2	-31.0	-31.8	0.2	-28.2	15.4	-2.1	13.6	-4.1	4.2	220	0.755

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	31.0	17.8	28.9	17.3	26.8	16.5	20.4	26.3	19.1	25.3	17.9	24.1	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.5	23.2	17.0	13.2	21.4	15.7	12.1	20.3	61.8	25.9	57.1	25.3	53.3	24.1	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.4	10.2	-34.3	34.8	2.7	1.8	-36.2	36.1	-37.7	37.2	-39.2	38.2	-41.2	39.6		
(5)			-34.4	23.1	2.6	2.0	-36.3	24.5	-37.8	25.6	-39.3	26.7	-41.2	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-12.4	-11.2	-4.5	4.3	10.9	15.5	18.5	17.6	12.9	5.1	-3.7	-10.5	(6)
(7)		DBStd	12.78	8.74	8.05	8.21	5.74	4.67	3.53	3.07	3.52	4.81	5.22	7.12	8.31	(7)
(8)		HDD10.0	3210	695	593	450	183	46	3	0	1	25	167	413	634	(8)
(9)		HDD18.3	5472	953	827	708	422	233	98	37	56	173	411	662	893	(9)
(10)		CDD10.0	878	0	0	1	11	74	168	262	238	111	13	1	0	(10)
(11)		CDD18.3	99	0	0	0	0	3	13	41	34	9	0	0	0	(11)
(12)		CDH23.3	1481	0	0	0	7	82	186	496	521	185	4	0	0	(12)
(13)		CDH26.7	461	0	0	0	1	16	46	153	186	59	0	0	0	(13)
(14)	Wind	WSAvg	5.1	5.4	5.1	5.3	5.5	5.6	5.2	4.3	4.2	4.8	5.3	5.3	5.2	(14)
(15)	Precipitation	PrecAvg	388	18	9	15	23	48	84	60	47	34	19	17	17	(15)
(16)		PrecMax	538	40	16	36	61	128	225	125	121	75	49	47	35	(16)
(17)		PrecMin	219	2	2	1	5	5	25	14	4	5	4	3	2	(17)
(18)		PrecStd	81	9	4	10	15	29	46	32	31	21	11	13	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	5.6	15.0	23.3	28.7	31.2	33.1	34.0	31.5	23.5	15.5	8.0	(19)
(20)			MCWB	3.2	2.7	8.4	11.0	14.3	17.4	19.1	17.4	16.1	12.9	8.4	3.9	(20)
(21)		2%	DB	3.5	3.3	11.2	19.7	25.6	27.6	30.4	31.2	28.2	19.0	10.5	5.0	(21)
(22)			MCWB	1.5	1.3	5.9	9.7	13.3	16.6	19.2	17.1	14.6	10.5	5.4	2.0	(22)
(23)		5%	DB	2.0	1.7	8.0	16.9	23.1	25.3	28.2	28.7	25.1	15.7	7.4	3.0	(23)
(24)			MCWB	0.4	0.2	4.1	8.1	12.0	15.3	18.9	16.8	13.8	9.0	3.7	0.8	(24)
(25)		10%	DB	-0.2	0.0	5.0	13.8	20.4	23.2	26.1	26.3	22.0	12.9	4.9	0.7	(25)
(26)			MCWB	-1.3	-1.1	2.6	6.9	10.9	14.7	17.8	16.2	12.6	7.6	2.4	-0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.1	8.7	12.2	16.0	20.6	23.3	20.2	17.5	13.7	8.6	4.1	(27)
(28)			MCDWB	5.5	5.3	14.6	21.0	23.5	26.0	28.7	27.4	27.8	21.0	15.0	7.2	(28)
(29)		2%	WB	1.7	1.5	6.2	10.3	14.2	18.3	21.5	19.2	15.7	11.3	5.9	2.2	(29)
(30)			MCDWB	3.3	3.1	10.4	17.9	22.8	24.2	27.1	26.7	25.2	17.4	9.9	4.7	(30)
(31)		5%	WB	0.4	0.3	4.2	8.9	13.0	16.9	20.0	18.2	14.5	9.5	4.1	0.9	(31)
(32)			MCDWB	1.9	1.7	7.4	15.8	20.9	22.7	25.6	25.5	23.0	14.6	7.0	3.0	(32)
(33)		10%	WB	-1.4	-1.1	2.6	7.4	11.8	15.7	18.7	17.1	13.3	8.0	2.5	-0.9	(33)
(34)			MCDWB	-0.2	0.0	5.0	13.0	18.9	21.2	24.1	23.9	21.2	12.5	4.7	0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	9.5	9.7	11.8	13.0	11.4	12.8	13.9	13.3	11.1	9.0	8.7	(35)
(36)			MCDBR	9.6	10.0	12.8	17.0	17.1	15.5	16.2	18.6	18.6	15.7	12.2	10.0	(36)
(37)			MCWBR	8.2	8.5	8.3	8.8	7.7	7.0	7.6	7.3	7.9	8.4	8.0	7.8	(37)
(38)		5% WB	MCDBR	10.0	9.9	12.0	15.7	15.0	12.6	14.0	15.0	16.8	14.3	11.5	10.1	(38)
(39)			MCWBR	8.7	8.5	8.1	8.6	7.3	6.7	7.8	6.9	7.6	8.2	7.8	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.253	0.250	0.271	0.337	0.355	0.356	0.364	0.373	0.328	0.283	0.253	0.224	(40)	
(41)		taud	2.336	2.400	2.500	2.353	2.352	2.404	2.395	2.338	2.445	2.554	2.494	2.423	(41)	
(42)		Ebn at Noon	792	907	948	904	900	900	885	856	865	858	802	794	(42)	
(43)		Edh at Noon	62	76	85	111	117	112	111	112	90	66	53	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.16	2.16	3.53	4.80	5.85	6.05	6.54	5.25	3.85	2.32	1.22	0.86	(44)	
(45)		RadStd	0.12	0.17	0.28	0.39	0.43	0.43	0.39	0.39	0.35	0.29	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page